

Phase 6A – Security Hardening Using Microsoft Azure and Azure Kubernetes Service (AKS)

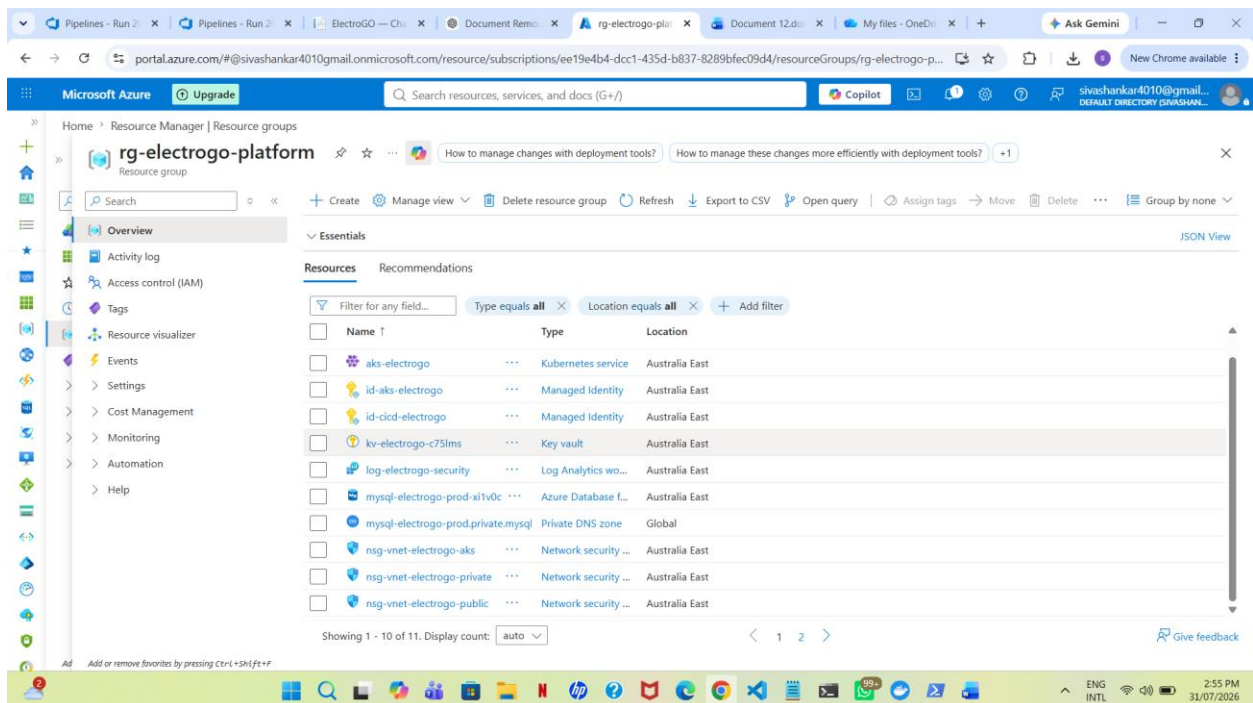
1. Project Aim

The primary objective of Phase 6A is to strengthen the security posture of the ElectroGo EV Charging Platform deployed on Microsoft Azure. This phase focuses on implementing enterprise-grade security controls to protect cloud infrastructure, Kubernetes workloads, application secrets, and user access.

Using Microsoft Azure security services and Kubernetes best practices, the platform adopts a **Defense-in-Depth** strategy, ensuring that security is enforced across every layer of the application—from the network edge to the application and data layers.

Screenshot Placement

Azure Resource Group showing all deployed security resources.



Name	Type	Location
aks-electrogo	Kubernetes service	Australia East
id-aks-electrogo	Managed Identity	Australia East
id-cid-electrogo	Managed Identity	Australia East
kv-electrogo-c75lms	Key vault	Australia East
log-electrogo-security	Log Analytics wo...	Australia East
mysql-electrogo-prod-xi1v0c	Azure Database f...	Australia East
mysql-electrogo-prod.private.mysql	Private DNS zone	Global
nsg-vnet-electrogo-aks	Network security ...	Australia East
nsg-vnet-electrogo-private	Network security ...	Australia East
nsg-vnet-electrogo-public	Network security ...	Australia East

2. Project Overview

The ElectroGo platform is a cloud-native EV charging application deployed on **Azure Kubernetes Service (AKS)**. Earlier phases focused on infrastructure provisioning, containerization, Kubernetes deployment, and monitoring. Phase 6A extends the project by implementing comprehensive security controls to safeguard infrastructure, applications, and sensitive data.

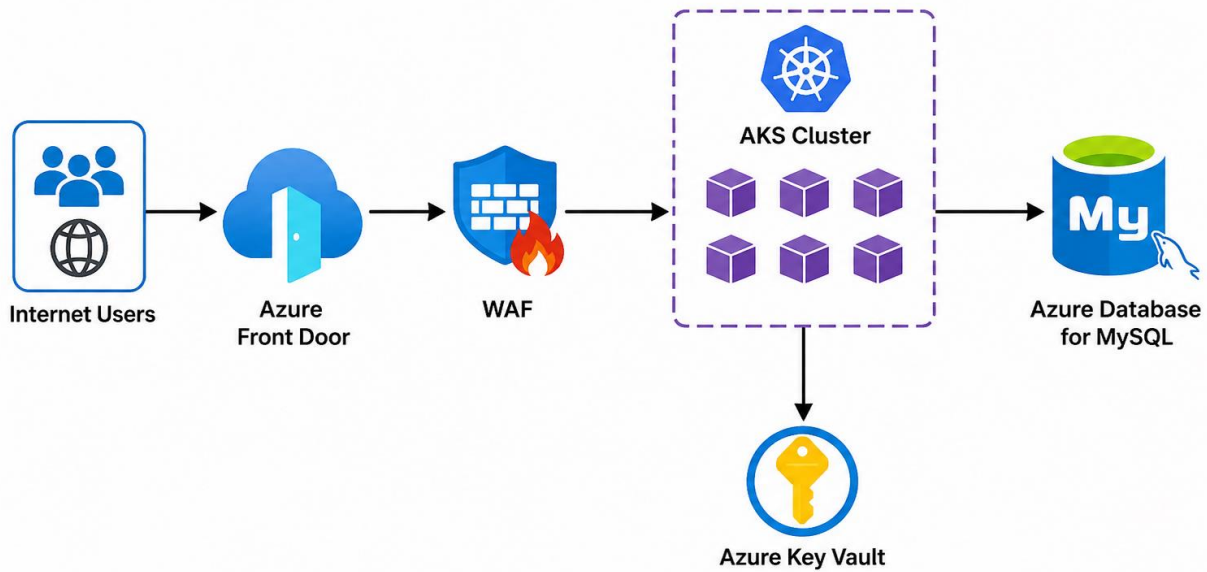
The solution integrates several Azure security services, including:

- Azure Key Vault
- Azure Front Door
- Azure Web Application Firewall (WAF)
- Azure Role-Based Access Control (RBAC)
- Microsoft Defender for Containers
- Azure Entra ID Authentication
- Kubernetes Network Policies
- HTTPS/TLS Encryption

Together, these services provide secure identity management, encrypted communication, vulnerability protection, and controlled access throughout the platform.

Screenshot Placement

Updated Solution Architecture showing Azure Front Door → WAF → AKS → Azure Database for MySQL → Azure Key Vault.



3. Project Objectives

The objectives of this phase include:

- Implement least-privilege access using Azure RBAC.
- Secure sensitive information using Azure Key Vault.
- Enable HTTPS communication using Azure-managed certificates.
- Protect web applications using Azure Front Door Web Application Firewall.
- Restrict pod-to-pod communication using Kubernetes Network Policies.
- Scan container images using Microsoft Defender for Containers.
- Secure application login using Microsoft Entra ID.
- Configure Azure AD RBAC for AKS cluster administration.
- Integrate security validation into the CI/CD deployment pipeline.

4. Security Architecture

The ElectroGo platform follows a layered security architecture to protect every component of the application.

Incoming user requests are first processed by **Azure Front Door**, where HTTPS encryption and Web Application Firewall policies inspect incoming traffic and block malicious requests before they reach the application.

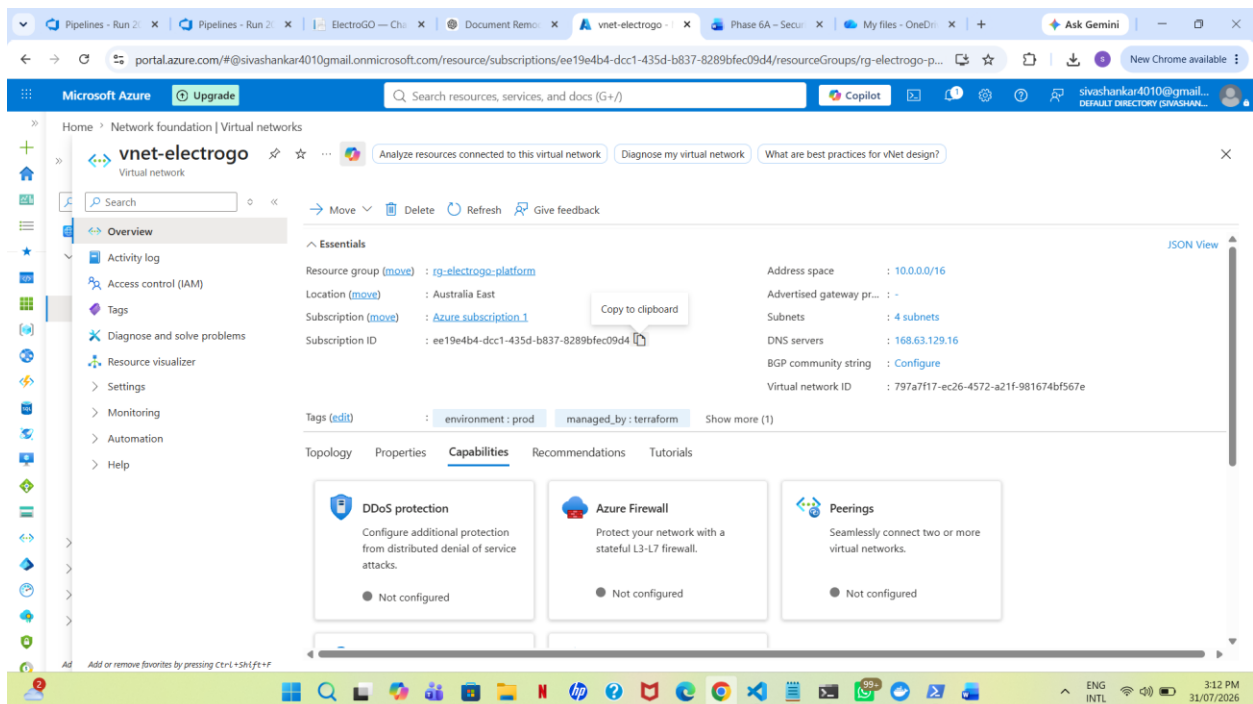
Validated requests are forwarded to the Kubernetes Ingress Controller running inside Azure Kubernetes Service. Application secrets are securely retrieved from Azure Key Vault using the Secrets Store CSI Driver, eliminating hardcoded credentials.

Azure RBAC controls resource access, Kubernetes Network Policies isolate workloads, and Microsoft Defender continuously monitors container images and runtime activity to detect vulnerabilities and security threats.

Screenshot Placement

Architecture Diagram

(VNet, Subnets, NSG, Front Door, AKS, MySQL, Key Vault)



Microsoft Azure | Subnets

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Name	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
snet-electrogo-agic	10.0.3.0/24	-	251	-	-	-
snet-electrogo-private	10.0.2.0/24	-	251	Microsoft...	nsg-vnet-e...	-
snet-electrogo-aks	10.0.16.0/20	-	4,033	-	nsg-vnet-e...	-
snet-electrogo-public	10.0.1.0/24	-	251	-	nsg-vnet-e...	-

Showing 4 subnets

Microsoft Azure | Network security groups

Check security rule conflicts across network security groups | Suggest a firewall SKU for my network topology

Name	Resource Group	Location	Subscription	Flow log
aks-agentpool-19414897-nsg	mc-rg-electrogo-platform...	Australia East	Azure subscription 1	
nsg-vnet-electrogo-aks	rg-electrogo-platform	Australia East	Azure subscription 1	
nsg-vnet-electrogo-private	rg-electrogo-platform	Australia East	Azure subscription 1	
nsg-vnet-electrogo-public	rg-electrogo-platform	Australia East	Azure subscription 1	

Showing 1 - 4 of 4. Display count: auto

The screenshot shows the 'aks-electrogo' Kubernetes service overview in the Azure portal. The left sidebar contains navigation links for Overview, Activity log, Access control (IAM), Tags, Monitor (Insights), Diagnose and solve problems, Microsoft Defender for Cloud, Cost analysis, Resource visualizer, and Kubernetes resources. The main content area is divided into sections: Essentials, Properties, Monitoring, and Recommendations. The Essentials section lists key properties such as Resource group (rg-electrogo-platform), Power state (Running), Cluster operation status (Succeeded), Subscription (Azure subscription 1), Location (Australia East), Subscription ID (ee19e4b4-dcc1-435d-b837-8289bfec09d4), and Fleet Manager (Click here to assign). The Properties section shows details for Kubernetes services (Encryption type, Encryption at-rest with a platform-managed key, Virtual node pools) and Networking (API server address, Network configuration, Pod CIDR, Service CIDR, DNS service IP). The bottom of the page shows a taskbar with various application icons and system status indicators.

The screenshot shows the 'Key vaults' overview in the Azure portal. The left sidebar contains navigation links for Key vaults, Create, Manage deleted vaults, Manage view, Refresh, Export to CSV, Open query, Assign tags, and Add to service group. The main content area displays a table of key vaults with the following columns: Name, Type, Resource Group, Location, Subscription, and Tags. The table lists two key vaults: 'electrogo-kv' and 'kv-electrogo-c75lms'. The bottom of the page shows a taskbar with various application icons and system status indicators.

Name	Type	Resource Group	Location	Subscription	Tags
electrogo-kv	Key vault	ElectroGo-RG	East US	Azure subscription 1	
kv-electrogo-c75lms	Key vault	rg-electrogo-platform	Australia East	Azure subscription 1	environment: prod

5. CI/CD Pipeline Security

The ElectroGo platform uses **Azure DevOps Pipelines** to automate application delivery from source code to Azure Kubernetes Service. The CI/CD pipeline follows DevSecOps

principles by integrating security validation before deployment, ensuring that only secure container images are released into production.

The deployment workflow consists of three automated stages:

Build

The pipeline restores application dependencies, compiles the ASP.NET Core application, builds the Docker container image, and pushes the image to Azure Container Registry (ACR).

Security Scan

The newly created container image is scanned using **Trivy**. The scan checks for operating system and application vulnerabilities. If any **High** or **Critical** severity vulnerabilities are detected, the pipeline automatically stops, preventing insecure images from reaching production.

Deploy

Once the security scan is successfully completed, Helm deploys the application to Azure Kubernetes Service. The deployment uses a namespace-scoped **OIDC-federated managed identity**, removing the need for stored credentials while following the principle of least privilege.

This automated deployment process improves deployment consistency, reduces manual intervention, and enforces security validation before every release.

Screenshot Placement

Azure DevOps Pipeline

(Build → Security Scan → DeployToAKS)

Jobs in run #2026073...

ElectroGo-Security-Implementation.git (4)

Build & Push Docker Image

BuildAndPush51s

Initialize job2s

Checkout ElectroGo...3s

Build and push ima...45s

Post-job: Checkout ...<1s

Finalize Job

View raw logView task inputs

1 ▶ Condition evaluation
9 Starting: Finalize Job
10 Cleaning up task key
11 Start cleaning up orphan processes.
12 Finishing: Finalize Job

Jobs in run #2026073...

ElectroGo-Security-Implementation.git (4)

Build & Push Docker Image

BuildAndPush51s

Trivy Image Scan Gate

TrivyScan43s

Initialize job2s

Checkout ElectroGo...1s

Trivy scan (with ign...38s

Post-job: Checkout ...<1s

Finalize Job<1s

Finalize Job

View raw logView task inputs

1 ▶ Condition evaluation
9 Starting: Finalize Job
10 Cleaning up task key
11 Start cleaning up orphan processes.
12 Finishing: Finalize Job

Jobs in run #20260731.5

Trivy Image Scan Gate

TrivyScan43s

Deploy to AKS (prod)

DeployHelm2m 27s

Initialize job3s

Checkout ElectroGo...2s

HelmInstaller1s

Get AKS credentials21s

Install NGINX Ingres...59s

Helm upgrade/insta...42s

Post-deploy verifica...15s

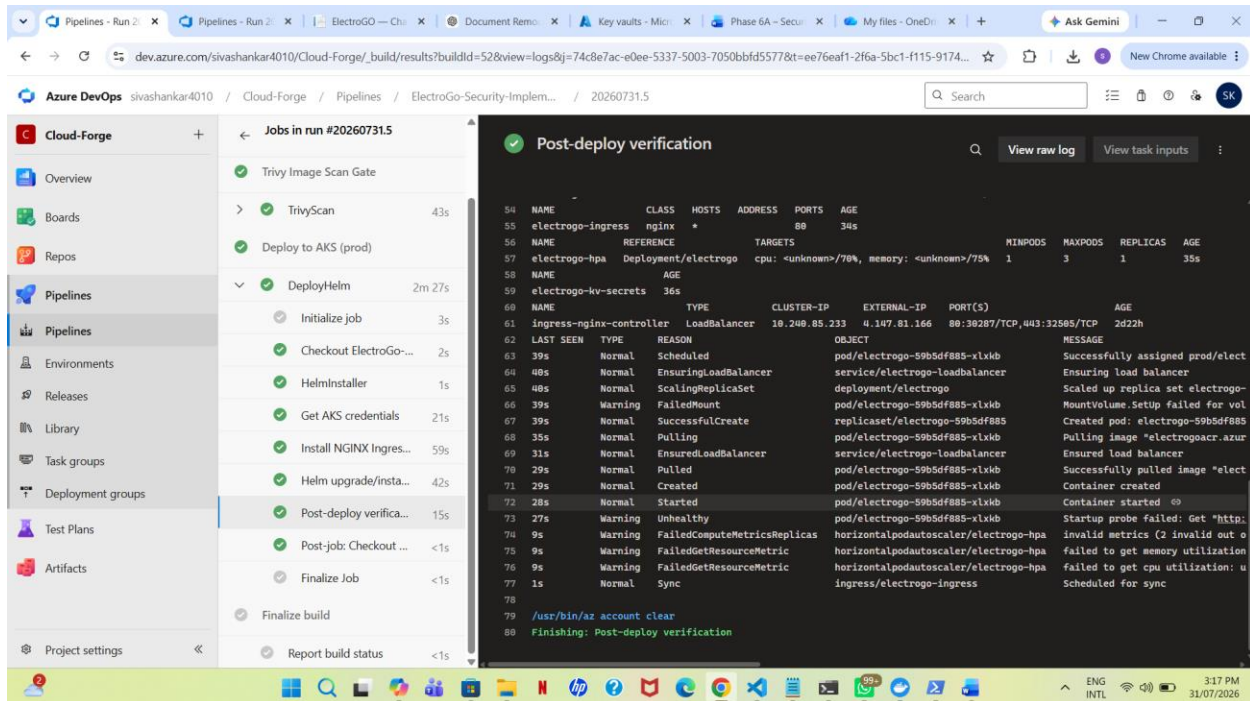
Post-job: Checkout ...<1s

Finalize Job<1s

Finalize Job

View raw logView task inputs

1 ▶ Condition evaluation
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11 Start cleaning up orphan processes.
12 Finishing: Finalize Job



6. Security Controls Implementation

Each security control was implemented to address a specific security requirement while improving the overall protection of the ElectroGo platform.

6.1 Azure RBAC

Azure Role-Based Access Control ensures that every user, service principal, and managed identity receives only the permissions required to perform assigned tasks. Separate identities are configured for administrators, Kubernetes services, and CI/CD pipelines.

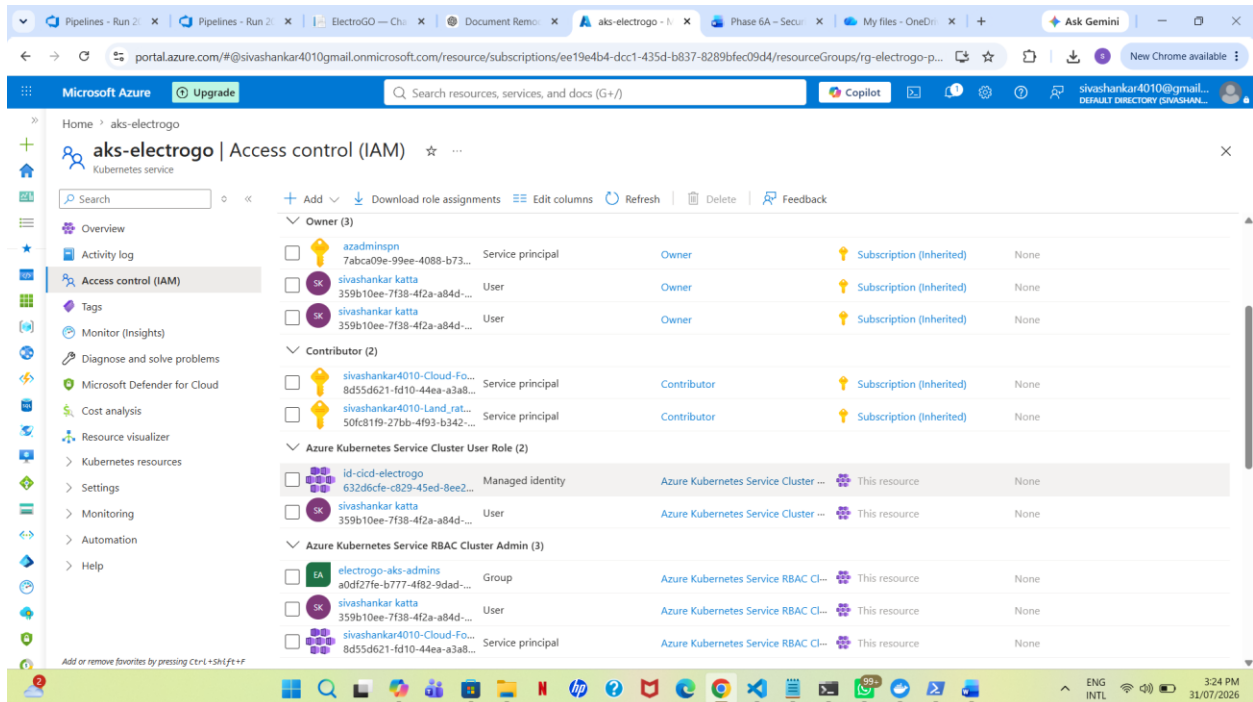
Benefits

- Least privilege access
- Improved identity management
- Reduced insider threats

Screenshot Placement

Before – Contributor Role

After – Custom RBAC Assignments



6.2 Azure Key Vault

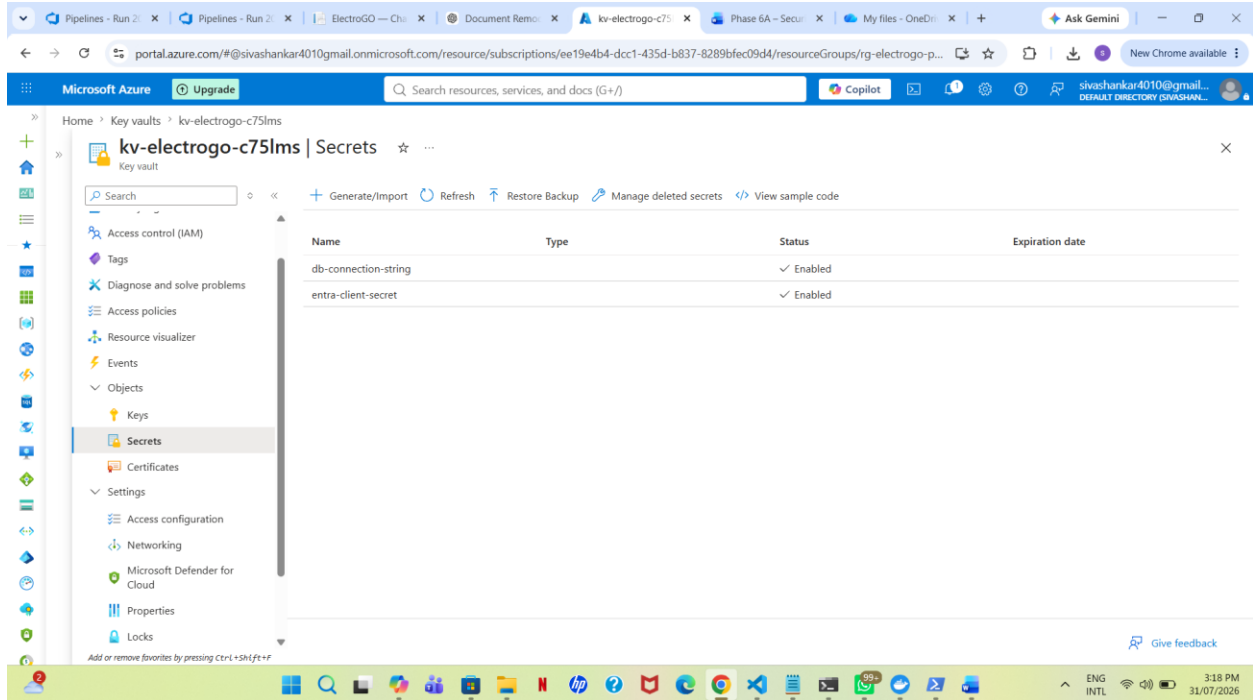
Azure Key Vault securely stores application secrets, connection strings, and sensitive configuration values. Kubernetes retrieves secrets dynamically using the Secrets Store CSI Driver, eliminating plaintext credentials.

Benefits

- Secure secret management
- No hardcoded credentials
- Improved compliance

Screenshot Placement

Key Vault Secret List



6.3 HTTPS and TLS

Azure Front Door provides HTTPS connectivity using Microsoft-managed certificates. All HTTP requests are automatically redirected to HTTPS, ensuring encrypted communication between users and the application.

Benefits

- Secure communication
- Data encryption
- Protection against interception attacks

6.4 Azure Front Door Web Application Firewall

Azure Front Door Web Application Firewall protects the application against common web attacks such as SQL Injection, Cross-Site Scripting (XSS), malicious bots, and other OWASP Top 10 vulnerabilities.

Benefits

- SQL Injection protection
- XSS protection
- Bot filtering
- Rate limiting

6.5 Kubernetes Network Policies

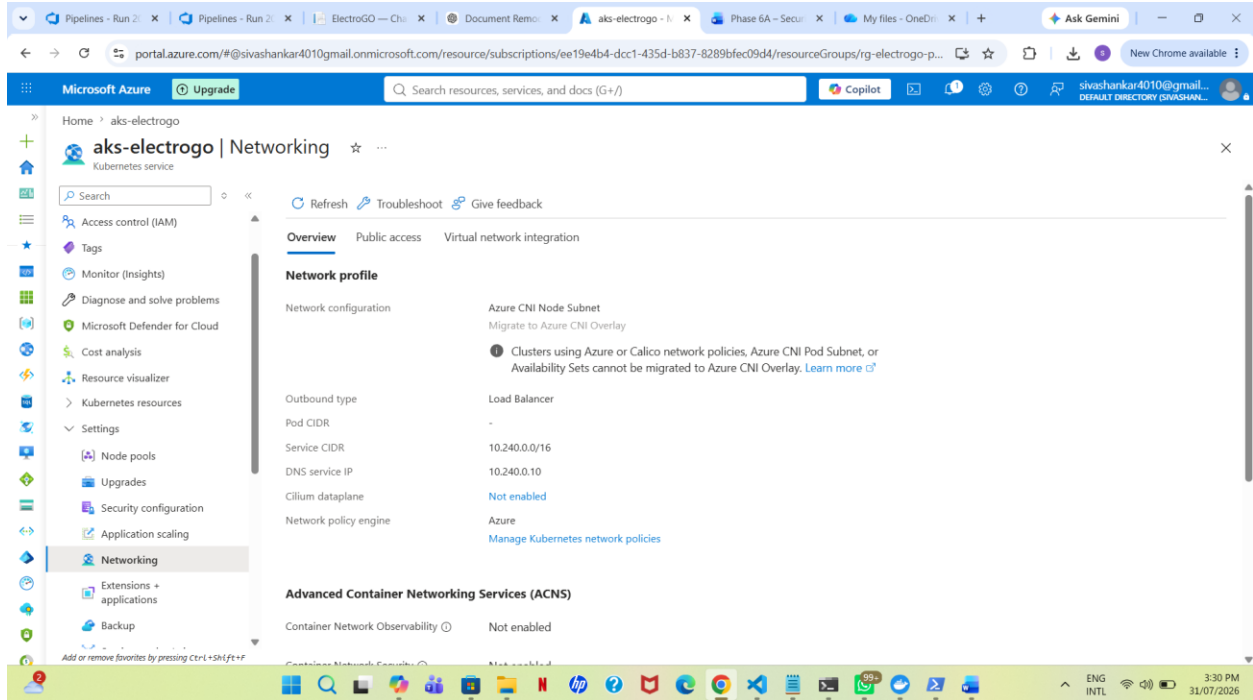
Network Policies implement a default-deny model that blocks unnecessary communication between Kubernetes pods. Only approved workloads are permitted to communicate.

Benefits

- Prevents lateral movement
- Improves workload isolation
- Reduces attack surface

Screenshot Placement

Network Policy Configuration



6.6 Microsoft Defender for Containers

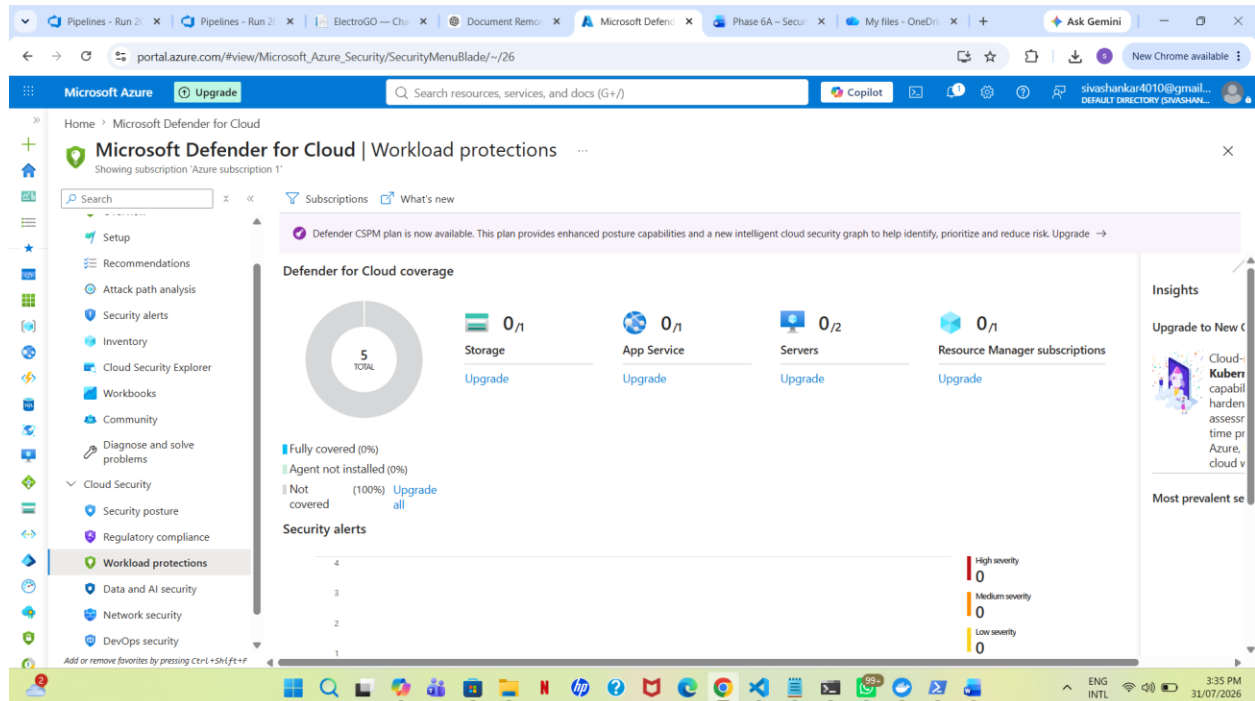
Microsoft Defender continuously scans Azure Container Registry images and monitors running Kubernetes workloads for vulnerabilities and runtime threats.

Benefits

- Vulnerability scanning
- Runtime protection
- Security recommendations

Screenshot Placement

Microsoft Defender Dashboard



6.7 Azure Entra ID Authentication

Administrative access is protected using Microsoft Entra ID, allowing only authenticated organizational users to access administrative functions.

Benefits

- Secure authentication
- Centralized identity management
- Conditional Access support

6.8 Azure AD RBAC for AKS

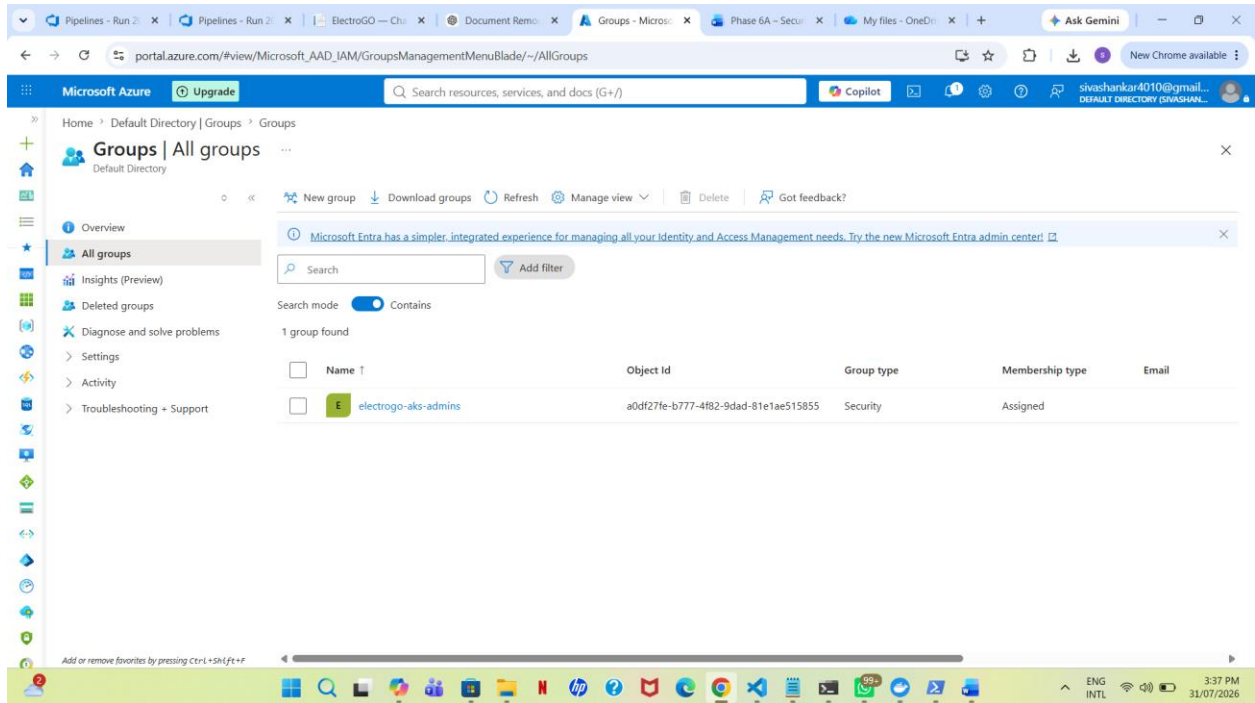
Azure AD-integrated RBAC controls Kubernetes cluster access through Azure AD groups. Permissions are assigned according to user roles, improving security and simplifying access management.

Benefits

- Secure cluster administration
- Group-based access control
- Centralized user management

Screenshot Placement

Azure AD Groups



The screenshot shows the Microsoft Azure portal interface for managing groups. The left sidebar contains navigation options like Overview, All groups, Insights (Preview), Deleted groups, Diagnose and solve problems, Settings, Activity, and Troubleshooting + Support. The main content area shows a list of groups, with one group 'electrogo-aks-admins' visible. The group is of type 'Security' and has an 'Assigned' membership type. The page includes a search bar, a filter button, and a table with columns for Name, Object Id, Group type, Membership type, and Email.

Name	Object Id	Group type	Membership type	Email
electrogo-aks-admins	a0df27fe-b777-4f82-9dad-81e1ae515855	Security	Assigned	

AKS RBAC Verification

Microsoft Azure | aks-electrogo | Access control (IAM)

Search resources, services, and docs (G+)

Download role assignments | Edit columns | Refresh | Delete | Feedback

Role	Principal	Contributor	Subscription (Inherited)	None
Subscription (Inherited)	8d55d621-fd10-44ea-a3a8...	Contributor	Subscription (Inherited)	None
Subscription (Inherited)	svashankar4010-Land_rat...	Contributor	Subscription (Inherited)	None
Azure Kubernetes Service Cluster User Role (2)				
id-cicd-electrogo-632d6cfe-c829-45ed-8ee2...	Managed identity	Azure Kubernetes Service Cluster ...	This resource	None
svashankar katta 359b10ee-7f3b-4f2a-a84d...	User	Azure Kubernetes Service Cluster ...	This resource	None
Azure Kubernetes Service RBAC Cluster Admin (3)				
electrogo-aks-admins-a0d27fe-b777-4f82-9dad...	Group	Azure Kubernetes Service RBAC Cl...	This resource	None
svashankar katta 359b10ee-7f3b-4f2a-a84d...	User	Azure Kubernetes Service RBAC Cl...	This resource	None
svashankar4010-Cloud-For...8d55d621-fd10-44ea-a3a8...	Service principal	Azure Kubernetes Service RBAC Cl...	This resource	None
Defender Agentless VM Scan (1)				
Microsoft Defender for Clo...0c7668b5-3260-4a00-9f53...	Service principal	Defender Agentless VM Scan	Subscription (Inherited)	None
Defender Containers Sensor (1)				
Containers/securityOperat...dc9a85e9-3db8-4e01-bb8f...	Managed identity	Defender Containers Sensor	Subscription (Inherited)	None
Defender For Container Registries Operator (1)				
Containers/securityOperat...				

ELECTRO GO

HOW IT WORKS | FEATURES | PRICING | NETWORK

LOGIN | REGISTER

THE FUTURE OF EV CHARGING

CHARGE SMARTER. GO FURTHER.

India's fastest growing EV charging network. 800+ stations, 10-minute rapid charge, and real-time availability — all in your pocket.

START CHARGING | EXPLORE NETWORK

7. Security Compliance Summary

Security Requirement	Status
Azure RBAC	✔ Implemented
Azure Key Vault	✔ Implemented
HTTPS/TLS Encryption	✔ Implemented
Azure Front Door WAF	✔ Implemented
Kubernetes Network Policies	✔ Implemented
Microsoft Defender for Containers	✔ Implemented
Azure Entra ID Authentication	✔ Implemented
AKS Azure AD RBAC	✔ Implemented
Secure CI/CD Pipeline	✔ Implemented

8. Challenges and Solutions

Challenge	Solution
Protecting sensitive application secrets	Azure Key Vault with CSI Driver
Preventing web attacks	Azure Front Door WAF
Restricting Kubernetes communication	Network Policies
Securing administrator access	Azure Entra ID Authentication
Detecting vulnerabilities	Microsoft Defender & Trivy
Secure automated deployments	Azure DevOps CI/CD with Security Scan

9. Conclusion

Phase 6A successfully enhances the security of the ElectroGo EV Charging Platform by implementing a comprehensive set of cloud-native security controls. Azure RBAC, Azure Key Vault, Azure Front Door, Web Application Firewall, Kubernetes Network Policies, Microsoft Defender for Containers, and Azure Entra ID work together to secure identities, applications, infrastructure, and sensitive data.

The integration of security scanning within the Azure DevOps CI/CD pipeline ensures that every deployment is validated before reaching production. By combining identity management, encrypted communication, workload isolation, vulnerability assessment, and automated security checks, the platform achieves a robust defense-in-depth architecture suitable for modern enterprise cloud applications.